



Road Traffic Analysis Using YOLOv4 and Deep SORT

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DOI: <https://doi.org/10.56025/IJARESM.140426361>

ABSTRACT

The rapid growth of urban populations and vehicular density has led to significant challenges in traffic management, including congestion, increased travel time, and environmental pollution. Traditional traffic monitoring systems rely on manual surveillance and sensor-based approaches, which are inefficient, costly, and lack scalability. To address these limitations, this research proposes an AI-powered road traffic analysis system using YOLOv4 for real-time vehicle detection and Deep SORT for multi-object tracking. The system processes live and recorded video streams from traffic cameras to detect and track vehicles such as cars, buses, trucks, and motorcycles. Each detected vehicle is assigned a unique ID, enabling accurate tracking across frames and preventing duplicate counting. Advanced preprocessing techniques, including frame extraction, normalization, and bounding box filtering, improve detection accuracy and performance. The system is implemented using Python, TensorFlow/PyTorch, OpenCV, and NumPy, ensuring efficient real-time processing. It provides insights into traffic density, vehicle movement patterns, and congestion hotspots. The proposed system enhances decision-making for traffic authorities, enabling optimized traffic flow, improved road safety, and support for smart city initiatives. Future work includes integration with IoT, predictive traffic analytics, and cloud-based deployment.

Keywords: Traffic Analysis, YOLOv4, Deep SORT, Vehicle Detection, Object Tracking, Computer Vision, Smart Cities, Deep Learning.

1. INTRODUCTION

Traffic congestion has become a critical issue in modern urban environments, resulting in delays, increased fuel consumption, and environmental pollution. Traditional traffic monitoring systems rely on manual observation, static sensors, and fixed signal timings, which are not capable of adapting to dynamic traffic conditions. Recent advancements in artificial intelligence and deep learning have introduced intelligent approaches for real-time traffic monitoring. The integration of YOLOv4 for object detection and Deep SORT for tracking provides a scalable and automated solution for traffic analysis. The proposed system processes video streams from traffic cameras to detect and track vehicles, analyze traffic patterns, and identify congestion hotspots. This enables authorities to make data-driven decisions for efficient traffic management and improved urban mobility.

1.1 Problem Statement

Existing traffic monitoring systems suffer from:

- Manual monitoring and human dependency
- High installation and maintenance cost
- Lack of real-time analysis
- Inability to handle high traffic density
- Limited accuracy in tracking and counting vehicles

There is a need for an intelligent system capable of automated, real-time traffic analysis using AI and deep learning.

1.2 Motivation

The motivation of this research is to develop an automated traffic monitoring system that:

- Reduces human effort
- Provides real-time traffic insights

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- Improves congestion management
- Enhances road safety
- Supports smart city infrastructure

1.3 Key objectives of this research include

The key objectives of this research are to design an AI-based traffic monitoring system using YOLOv4 for real-time vehicle detection and Deep SORT for accurate multi-object tracking; to analyze traffic density, vehicle count, and movement patterns using video data; to develop a scalable system capable of processing live and recorded video streams; to improve traffic management through data-driven insights; and to provide an efficient, cost-effective alternative to traditional traffic monitoring systems.

2. LITERATURE SURVEY

Recent advancements in computer vision and deep learning have significantly transformed intelligent transportation systems by enabling automated traffic monitoring, vehicle detection, and congestion analysis. Traditional traffic analysis methods relied on manual counting and sensor-based systems, which are inefficient and prone to errors. Modern approaches leverage deep learning models such as YOLO (You Only Look Once) for real-time object detection and Deep SORT for multi-object tracking. These methods improve accuracy, scalability, and real-time performance in traffic surveillance applications. The following table summarizes key contributions from existing research works relevant to the proposed system.

S.No	Citation	Research Focus	Methodology	Key Findings
1	Kusumah et al., 2023	Vehicle detection & counting	YOLOv4 + Deep SORT	Achieved efficient vehicle counting with tracking accuracy improvements
2	Zhang et al., 2022	Vehicle detection	Improved YOLOv5	Reduced false detection using Flip-Mosaic augmentation
3	Lin et al., 2021	Traffic monitoring	YOLO + GMM	Improved vehicle counting and speed estimation accuracy
4	Azimjonov et al., 2021	Vehicle detection & tracking	YOLO + BBox tracking	Increased detection accuracy up to 95% using hybrid models
5	Shubho et al., 2022	Traffic violation detection	YOLOv4 + OpenCV	Enabled real-time detection of traffic violations
6	Luo et al., 2024	Traffic monitoring	YOLOv5 + Deep SORT	Improved tracking stability and speed estimation
7	Vohra et al., 2023	Traffic monitoring	YOLO-based detection	Demonstrated efficiency of YOLO in real-time vehicle detection
8	Kunekar et al., 2024	Traffic signal optimization	YOLO + ML	Improved traffic flow using density-based signal control
9	Mandal et al., 2020	Vehicle counting	YOLO + Deep SORT	Achieved high accuracy in vehicle counting under varying conditions
10	Rezaei et al., 2021	Intelligent traffic systems	YOLO + SORT	Enabled 3D traffic monitoring and congestion detection

3. BACKGROUND WORK

The development of intelligent traffic monitoring systems has evolved significantly with advancements in computer vision, deep learning, and real-time analytics. Traditional traffic systems relied on manual observation and sensor-based infrastructure, which are limited in scalability and accuracy. The introduction of deep learning-based object detection and tracking models has enabled automated, accurate, and real-time traffic analysis.

3.1 Object Detection using YOLOv4

YOLOv4 (You Only Look Once version 4) is a real-time object detection model known for its speed and accuracy. Unlike traditional region-based detectors, YOLO processes the entire image in a single forward pass, making it highly efficient for real-time applications.

Key advantages:

- High detection speed (real-time performance)
- Ability to detect multiple objects simultaneously
- Robust performance under varying lighting conditions

In traffic analysis, YOLOv4 is used to detect different vehicle classes such as cars, buses, trucks, and motorcycles.

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3.2 Multi-Object Tracking using Deep SORT

Deep SORT (Simple Online and Realtime Tracking with Deep Learning) extends the SORT algorithm by incorporating appearance features for improved tracking.

Key features:

- Assigns unique IDs to each vehicle
 - Maintains identity across frames
 - Prevents duplicate counting
- This ensures accurate vehicle tracking even in crowded traffic scenarios.

3.3 Video Processing and Computer Vision Techniques

The system utilizes OpenCV and NumPy for handling video streams and preprocessing operations:

- Frame extraction from video
- Image resizing and normalization
- Noise reduction
- Bounding box refinement

These preprocessing steps improve detection accuracy and system performance.

3.4 Traffic Analytics and Pattern Recognition

Traffic analytics involves extracting meaningful insights from vehicle movement data, including:

- Vehicle density estimation
- Traffic flow patterns
- Congestion detection
- Peak-hour analysis

These analytics support intelligent traffic management and decision-making.

3.5 Intelligent Transportation Systems (ITS)

The integration of AI-based traffic analysis into ITS enables:

- Smart traffic signal control
- Automated congestion monitoring
- Real-time traffic alerts
- Data-driven urban planning

4. PROPOSED MODEL

The proposed system is an intelligent real-time road traffic analysis framework designed to automatically detect, track, and analyze vehicles using deep learning techniques. It integrates YOLOv4 for object detection and Deep SORT for multi-object tracking to ensure accurate identification and monitoring of vehicles across video frames. The system processes live or recorded video feeds and extracts meaningful insights such as traffic density and congestion patterns. The modular design ensures scalability, real-time performance, and adaptability for smart city applications.

4.1 Input Acquisition

The input acquisition module is responsible for capturing traffic video data either from live surveillance cameras or pre-recorded video files. It supports multiple video formats and ensures continuous frame streaming for processing. This module acts as the entry point of the system, providing raw visual data for further analysis. The quality and resolution of input data significantly influence detection accuracy and system performance.

4.2 Preprocessing

The pre-processing module prepares the input video frames for efficient analysis by applying several image processing techniques. It includes frame extraction, resizing, normalization, and noise reduction to improve image quality. These steps ensure consistency in input data and enhance the performance of the detection model. Proper preprocessing reduces computational complexity and increases the robustness of the system under varying lighting and environmental conditions.

4.3 Vehicle Detection (YOLOv4)

The vehicle detection module uses the YOLOv4 deep learning model to identify different types of vehicles in each frame. It generates bounding boxes around detected objects along with confidence scores indicating prediction accuracy. YOLOv4 processes images in a single pass, enabling real-time detection with high speed and precision. This module forms the core of the system by accurately identifying vehicles such as cars, buses, trucks, and motorcycles.

4.4 Vehicle Tracking (Deep SORT)

The tracking module utilizes the Deep SORT algorithm to monitor vehicles across consecutive frames. It assigns a unique ID to each detected vehicle and maintains its identity throughout the video sequence. By combining motion and appearance features, Deep SORT ensures accurate tracking even in crowded scenarios. This module prevents duplicate counting and enables analysis of vehicle movement patterns.

4.5 Traffic Analysis Engine

The traffic analysis module processes detection and tracking data to generate meaningful insights. It calculates vehicle counts, estimates traffic density, and identifies congestion zones in real time. The module also analyzes movement patterns and detects peak traffic conditions. These insights help in understanding traffic behavior and support intelligent decision-making for traffic management.

4.6 Data Management

The data management module is responsible for storing and organizing processed data, including video frames, detection results, and tracking information. It ensures efficient data handling and retrieval for analysis and reporting. This module also maintains historical records, enabling trend analysis and performance evaluation over time. Proper data management enhances system scalability and reliability.

4.7 Visualization Dashboard

The visualization module presents the output in an interactive and user-friendly format. It displays bounding boxes around detected vehicles, shows real-time vehicle counts, and provides graphical insights such as traffic density and congestion levels. The dashboard helps users easily interpret results and supports monitoring of traffic conditions. It enhances user experience and enables effective decision-making.

4.8 Parallel Processing

The system supports parallel processing to handle multiple tasks simultaneously, such as detection, tracking, and analysis. This ensures real-time performance even when processing high-resolution video streams. Parallel execution improves efficiency and reduces latency, making the system suitable for real-world deployment in busy traffic environments.

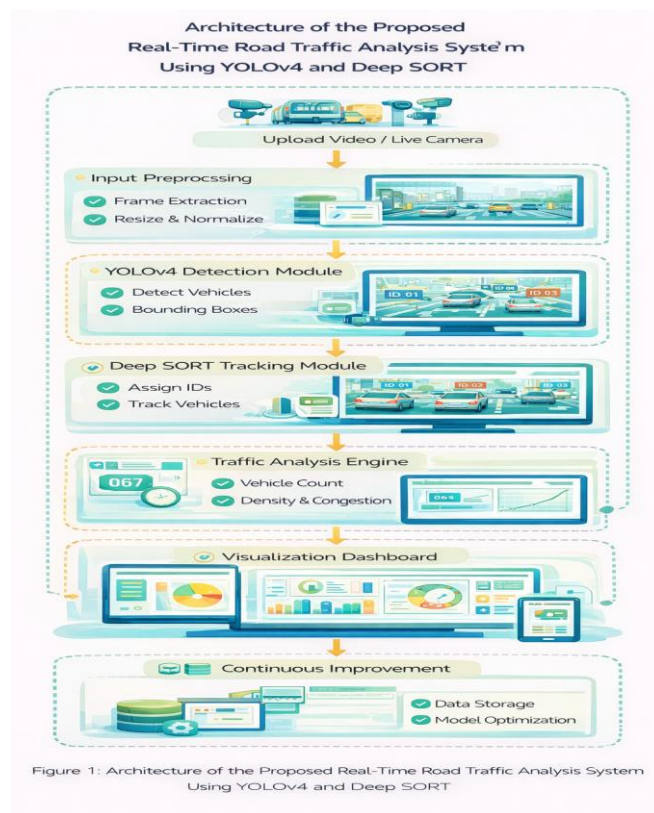


Figure 1: Architecture of the Proposed Real-Time Road Traffic Analysis System Using YOLOv4 and Deep SORT

Figure 1 illustrates the overall architecture of the proposed real-time road traffic analysis system designed for intelligent vehicle detection, tracking, and congestion monitoring. The system begins with video input acquisition, where traffic data is captured through live surveillance cameras or uploaded video streams. The input is then processed

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in the preprocessing stage, which includes frame extraction, resizing, and normalization to enhance image quality and ensure consistency.

The processed frames are passed to the YOLOv4 detection module, where vehicles such as cars, buses, and trucks are identified using bounding boxes and confidence scores. Following detection, the Deep SORT tracking module assigns unique IDs to each vehicle and tracks their movement across frames, ensuring accurate counting and preventing duplication.

5. IMPLEMENTATION RESULTS

The XOJ system was implemented as a complete web-based application integrating machine learning, explainable AI, and visualization tools.

System Implementation Overview

The implementation includes:

- Backend development using Flask
 - Frontend design using modern UI frameworks
 - Integration of ML models and XAI tools
- The modular structure ensures scalability and maintainability.

1) Login Page:

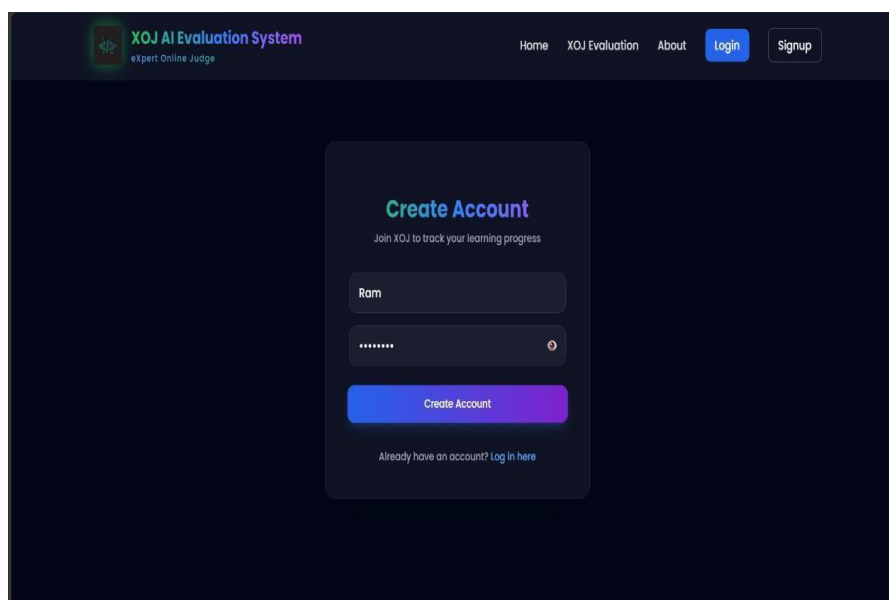


Figure 2: Create Account page where a new user registers by entering required details.

Figure 2 explains about the initial access point of the system that allows authorized users to securely enter the application. This interface typically contains input fields for the username and password along with a login button to authenticate the user. It ensures that only valid users can access the emotion detection system and its features.

2) Student Analyzer

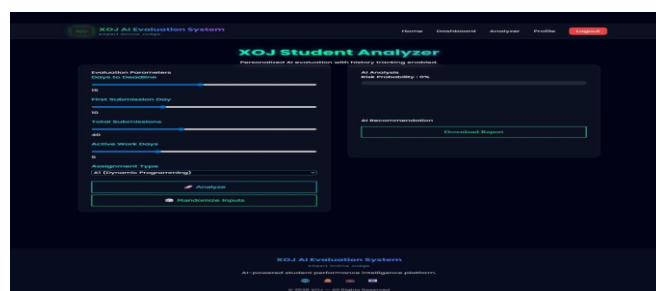


Figure 3. XOJ Student Analyzer Interface

Figure 3 explains the interface where the system takes user input and try to analyze the student performance.

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3) Results Page

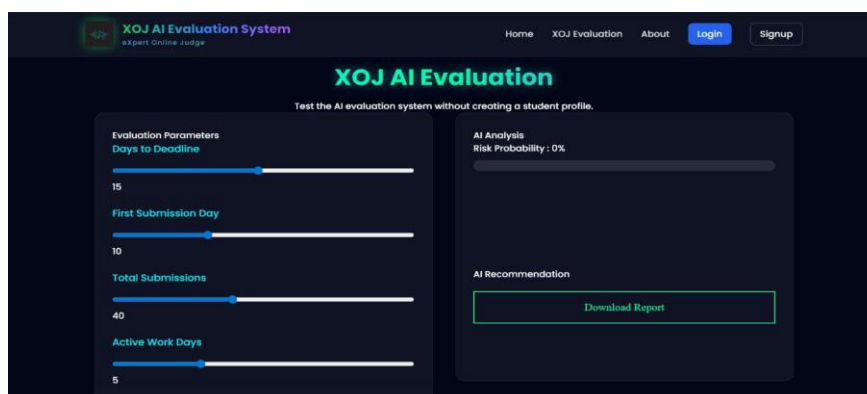


Figure 4. Analyze Student Performance without creating an account

Figure 4 displays the system displays feature contribution visualizations to explain the prediction outcome.

CONCLUSION

The proposed Student Performance Prediction System for Online Judge platforms effectively demonstrate the integration of machine learning and explainable artificial intelligence to enhance academic evaluation processes. By incorporating behavioral learning analytics such as submission patterns, consistency, and engagement levels, the system moves beyond traditional score-based assessment and provides a more comprehensive understanding of student performance. The implementation of the Random Forest algorithm ensures reliable and accurate predictions, while the integration of SHAP-based explainable AI enhances transparency by clearly identifying the contribution of each feature to the final prediction. This interpretability enables educators to gain deeper insights into student learning behavior and supports informed decision-making for early intervention. The system's web-based architecture, built using Flask and SQLite, ensures scalability, usability, and efficient data management. Visualization tools such as dashboards, confusion matrices, and performance graphs further improve user interaction and understanding of analytical results. Overall, the proposed framework provides a robust, scalable, and interpretable solution for predicting student performance in online judge environments. It contributes significantly to the advancement of educational data mining and intelligent learning systems. Future enhancements can focus on real-time analytics, integration with learning management systems, and the use of advanced deep learning models to further improve prediction accuracy and system adaptability.

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